



ONLINE AUCTION SYSTEM



A PROJECT REPORT

Submitted by

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*in partial fulfillment of the requirements for the award degree of
Bachelor of Technology*

CGB1221 - DATABASE MANAGEMENT SYSTEMS

DEPARTMENT OF INFORMATION TECHNOLOGY

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM - 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled “**ONLINE AUCTION SYSTEM**” has been carried out by the students ANJALI A, ASHWINI R, CHERMA SELVI A, DEEPANETHRA A. The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The Online Auction System is a web-based application designed to simplify the process of buying and selling products through auctions. The system provides a user-friendly interface where users can register, log in, and participate in auctions either as buyers or sellers. Sellers can list items with a starting price, while buyers can place bids on available products. The system includes features such as product listing, real-time bidding, auction time management, and automatic winner selection based on the highest bid. It also provides secure user authentication to ensure safe access and transactions. The bidding module allows users to track current bids and make competitive offers easily. One of the key features of this system is auction management, which controls the start and end time of auctions and ensures fair bidding. The system also maintains records of users, items, and bids using a database for efficient data handling. Overall, the system demonstrates the practical implementation of web technologies in creating an efficient, transparent, and interactive online auction platform.

Keywords:

Online Auction System, E-Commerce, PHP, MySQL, Bidding System, Auction Management, User Authentication.

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SIGNATURE

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LIST OF SYMBOLS AND ABBREVIATIONS

DBMS	–	DATABASE MANAGEMENT SYSTEM
SQL	–	STRUCTURED QUERY LANGUAGE
UI	–	USER INTERFACE
UX	–	USER EXPERIENCE
PHP	–	HYPERTEXT PREPROCESSOR
HTTP	–	HYPERTEXT TRANSFER PROTOCOL
CSS	–	CASCADING STYLE SHEETS
URL	–	UNIFORM RESOURCE LOCATOR
API	–	APPLICATION PROGRAMMING INTERFACE

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

An Online Auction System is an important application in today's digital world, enabling buyers and sellers to participate in auctions from anywhere at any time. It provides a platform where users can list items for sale and others can place bids, making the buying and selling process more dynamic and competitive. Traditionally, auctions were conducted physically, which limited participation and required more time, effort, and resources. This system automates the entire auction process, including user registration, item listing, bidding, and winner selection. It allows users to easily view available items, place bids in real-time, and track auction results. The system reduces manual work and paperwork while improving speed and accuracy. It ensures fair competition by maintaining transparency in bidding and preventing manipulation. Overall, the Online Auction System improves efficiency, accessibility, and reliability, making it a valuable solution for modern e-commerce and digital marketplaces.

1.1 PROBLEM DEFINITION

Traditional auction systems require people to be physically present, which makes the process time-consuming and inconvenient. These systems have limited reach, as only people in a specific location can participate. Managing auctions manually can lead to errors in recording bids and handling participants. There is also a lack of transparency, which may result in unfair practices. Tracking bidder information and maintaining records becomes difficult without a proper system. Communication between buyers and sellers is also limited in traditional methods. Handling multiple auctions at the same time

is not easy. The system ensures transparency by displaying real-time bid updates. It also maintains accurate records of users and transactions. Multiple auctions can be managed easily at the same time. Users can receive instant notifications about their bids. The system improves efficiency and saves time for both buyers and sellers. It also ensures better security and reliability. Overall, it provides a convenient and modern approach to auctions

1.3 OBJECTIVES OF THE PROJECT

The main objective of the Online Auction System is to provide a platform for buying and selling items through online bidding. It aims to eliminate the need for physical presence in traditional auctions. The system helps users to easily register and participate in auctions. It allows sellers to list their products with proper details and pricing. Buyers can place bids in a simple and convenient way. The system ensures transparency by showing real-time bid updates. It reduces manual errors by automating the auction process. It maintains proper records of users, items, and bids. The system aims to provide security for user data and transactions. It supports multiple auctions at the same time. Notifications are provided to keep users updated about auction status. The system improves efficiency and saves time for users. It provides a user-friendly interface for easy access. It helps in fair and accurate bidding. Overall, the objective is to create a reliable, fast, and secure auction system.

1.4 SCOPE OF THE PROJECT

The scope of the Online Auction System includes providing an efficient platform for online buying and selling of items through bidding. It allows users to register and log in securely to access the system. Sellers can list their products with details such as price and auction duration. Buyers can browse available items and place bids easily. The system updates bids in real time and shows the highest bid to all users. It manages multiple auctions at the same time without confusion.

It provides a user-friendly interface for easy usage. It also ensures security of user data and transactions. Overall, it covers all essential features needed for a smooth and reliable auction process.

1.5 DATABASE MANAGEMENT SYSTEM CONCEPT

A Database Management System (DBMS) is used in the Online Auction System to store, manage, and retrieve data efficiently. It plays a crucial role in handling large volumes of data related to customers, products, orders, and payments. By using DBMS concepts, the system ensures data consistency, security, and reliability while supporting smooth system operations.

- **Database Design:** Structured tables are created for users, products, orders, payments, and delivery details.
- **Normalization:** Data is normalized to reduce redundancy and avoid data anomalies.
- **Primary Key:** Used to uniquely identify each record in a table.
- **Foreign Key:** Used to establish relationships between tables.
- **CRUD Operations:** Create, Read, Update, and Delete operations are performed on database records.

CHAPTER 2

LITERATURE SURVEY

The literature survey of the Online Auction System focuses on studying existing auction methods and related technologies. Traditional auction systems were analyzed to understand their limitations such as time consumption and lack of transparency. Various online platforms were reviewed to identify features like real-time bidding, user authentication, and secure transactions. The survey helps in understanding current trends and technologies used in auction systems. Overall, it provides a foundation for developing a better and more efficient Online Auction System.

2.1 EXISTING SYSTEM

The existing system of auctions is mostly based on traditional methods where buyers and sellers must be physically present at a specific location. This process is time-consuming and limits participation to a small group of people. Managing bids manually can lead to errors and lack of transparency. There is no proper system to maintain records of users, items, and bids efficiently. Communication between participants is also limit. Security of transactions and data is not well maintained. Handling multiple auctions at the same time is difficult. Hence, the existing system is less efficient and not convenient for modern users.

2.2 LIMITATIONS OF EXISTING SYSTEM

The existing auction system has many limitations that reduce its efficiency and usability. It requires users to be physically present, which is time-consuming and inconvenient. The system has a limited reach, allowing only a small number of participants. Managing auctions manually can lead to errors in

recording bids. There is often a lack of transparency in the bidding process.. There may be delays in updating bid information and announcing results. Security of transactions and user data is not well ensured. The process is not user-friendly and requires more effort. It also increases the chances of human mistakes. Overall, the system is slow and less reliable. Hence, it is not suitable for modern needs.

2.4 PROPOSED SYSTEM

The proposed system is an Online Auction System that provides a simple and efficient way to conduct auctions. It allows users to participate from anywhere using the internet. Users can register and log in securely to access the system. Sellers can list their products with details and pricing. Buyers can view items and place bids easily. The system updates bids in real time to ensure transparency. It automatically selects the highest bidder as the winner. The system maintains proper records of users and transactions. It reduces manual work and improves accuracy. Overall, it is a fast, secure, and user-friendly solution for auctions.

2.4 ADVANTAGES OF PROPOSED SYSTEM

The proposed system offers many advantages over the existing system. It allows users to participate in auctions from anywhere at any time. It saves time and reduces the need for physical presence. The system provides real-time bid updates, ensuring transparency. It reduces manual errors by automating the auction process. It supports multiple auctions at the same time efficiently. The system maintains accurate records of users and transactions. It ensures better security for user data and payments. It provides a user-friendly interface for easy access. Overall, the system is fast, reliable, and convenient to use.

1.EasyAccess

Users can participate in auctions anytime and from anywhere using the internet.

2.TimeSaving

No need for physical presence → auctions happen quickly online.

3.UserFriendly

Simple interface makes it easy for buyers and sellers to use.

4.Real-TimeBidding

Users can place bids instantly and see updates immediately.

5.ReducedPaperwork

All records are stored digitally → no manual work.

6.HighTransparency

Bidding process is clear and fair → reduces cheating.

7.BetterDataManagement

Stores user details, items, and bids in an organized database.

8.SecureTransactions

User data and transactions are protected using security measures.

CHAPTER 3

SYSTEM DESIGN

3.1 ER DIAGRAM

An ER (Entity-Relationship) Diagram is used to represent the structure of the Online Auction System database. It shows the main entities such as User, Product, Auction, and Bid. Each entity contains attributes like user details, product information, and bid amount. The diagram also shows relationships between entities, such as users placing bids and sellers listing products. It helps in understanding how data is organized and connected. The ER diagram makes database design simple and clear. Overall, it provides a visual representation of the system structure.

ER DIAGRAM FOR ONLINE AUCTION SYSTEM

The ER diagram contains three main entities: Users, Items, and Bids.

- **Users Table:** This stores user details like *user_id*, *name*, *email*, *password*, and *created_at*. Each user has a unique ID, and email is also unique.
- **Items Table:** This stores product details such as *item_id*, *title*, *starting_price*, *image*, *category*, and *created_at*. The field *added_by* (FK) links the item to the user who added it.
- **Bids Table:** This stores bidding details like *bid_id*, *item_id*, *user_id*, *bid_amount*, and *created_at*. It connects both users and items.

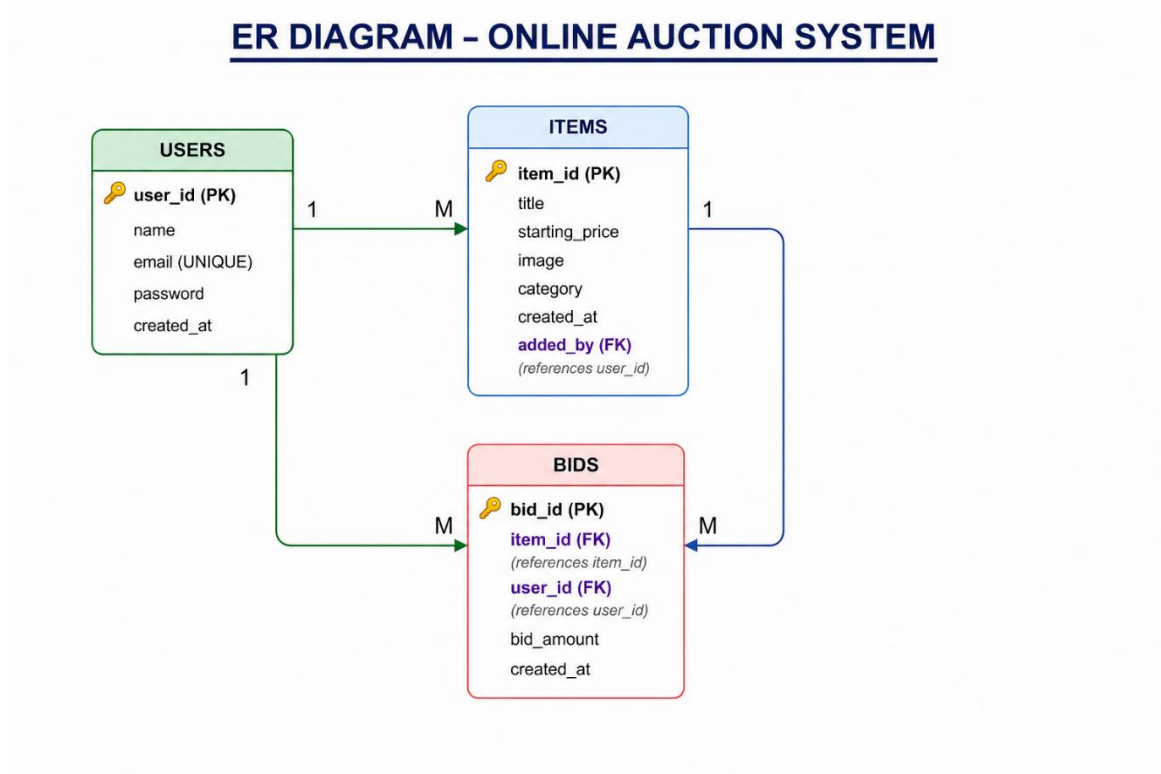


Fig 3.1 ER Diagram

3.2 DATABASE DESIGN

Database design in the Online Auction System defines how data is organized and stored efficiently. It includes creating tables such as User, Product, Auction, and Bid with their respective attributes. Each table is connected using relationships like primary keys and foreign keys. The design ensures data consistency and reduces redundancy through normalization. It allows easy insertion, updating, and retrieval of data. Proper database design also improves system performance and security. Overall, it helps in managing auction data in an organized and reliable manner.

3.2.1 Table Structure

The table structure in the Online Auction System defines how data is stored in different tables. Each table contains rows and columns representing records and attributes. Common tables include User, Product, Auction, and Bid. Each table has a primary key to uniquely identify records. Foreign keys are used to create relationships between tables. The structure ensures data is organized and easy to access. Overall, it helps in efficient storage and management of auction data.

3.2.2 Attributes

Each table contains attributes (fields) that describe the data stored in it. For example:

- User → User_ID, Name, Email, Password
- Product → ProductID, ProductName, Description, Starting_Price
- Auction → Auction_ID, Start_time, End_time, Status
- Bid → Bid_ID, Bid_Amount, Time

These attributes define the properties of each entity in the system.

3.4.3 Keys (Primary And Foreign Key)

Keys are used to uniquely identify records and establish relationships between tables.

- **Primary Key:** A unique identifier for each record in a table (e.g., UserID, ProductID).
- **Foreign Key:** A field that links one table to another (e.g., UserID in Order table, ProductID in Product table).

Primary and foreign keys ensure data integrity and help in maintaining relationships between different tables in the database.

CHAPTER 4

PROJECT METHODOLOGY

4.1 WORKFLOW OF THE SYSTEM

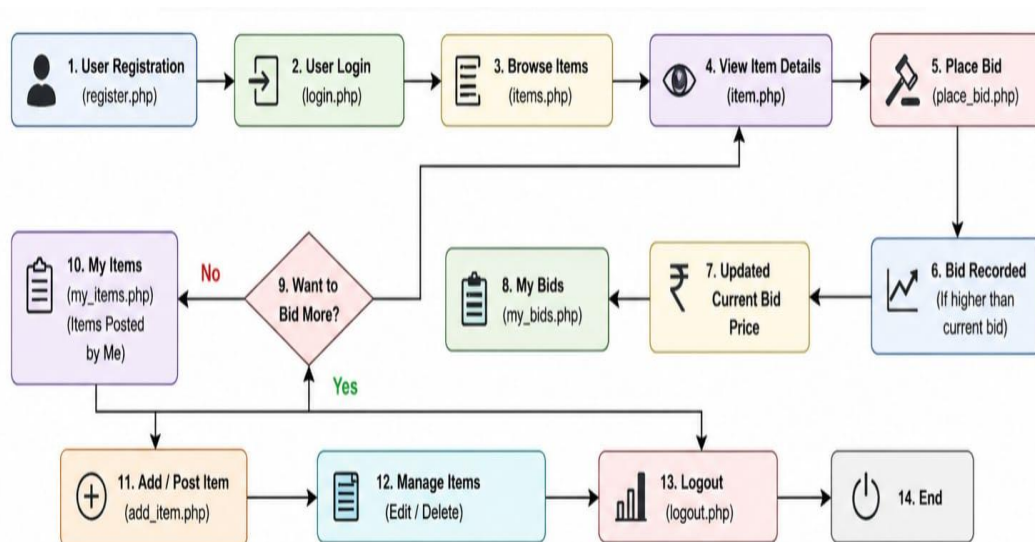


Fig 4.1 System Workflow

4.2 BLOCK DIAGRAM

BLOCK DIAGRAM – ONLINE AUCTION SYSTEM

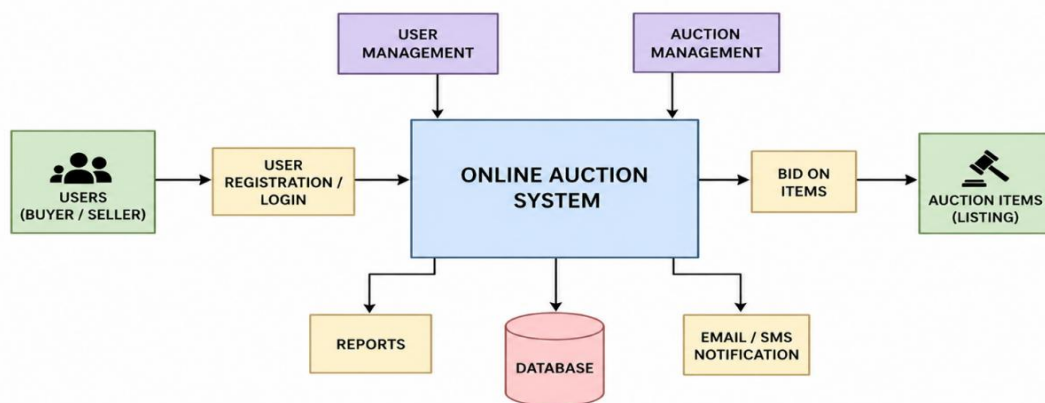


Fig 4.1 System Workflow

CHAPTER 5

SYSTEM REQUIREMENTS

The system requirements define the necessary hardware and software needed to develop and run the Online Auction System. Since the system is database-driven, special emphasis is given to database support and performance.

5.1. HARDWARE REQUIREMENTS

The hardware requirements ensure smooth functioning of the database and application.

Minimum Requirements:

1. Computer / Laptop
2. Processor (Intel i3 or above)
3. Minimum 4GB RAM
4. Hard Disk (at least 500GB)
5. Keyboard and Mouse
6. Stable Internet Connection

The database server requires higher RAM and storage because it handles large volumes of data such as user details, product inventory, and transaction records.

5.2 SOFTWARE REQUIREMENTS

This system is built using database-supported technologies for efficient data storage and retrieval.

1. Operating System (Windows / Linux)
2. Web Browser (Chrome, Edge, Firefox)
3. Programming Language (Java / Python / PHP)
4. Database (MySQL)
5. Web Server (Apache / XAMPP)
6. Development Tools (VS Code / Eclipse)

5.3 FUNCTIONAL REQUIREMENTS

- User Registration and Login
- Add / Manage Auction Items
- View Available Products
- Place Bids on Items
- Real-time Bid Updates
- Auction Start and End Time Management
- Automatic Winner Selection
- Admin Control (Manage users, items, bids)
- Search and Filter Items
- Notification to Users

CHAPTER 6

MODULE DESCRIPTION

6.1 USER MODULE

The User Module provides an interface for general users, such as buyers and sellers, to interact with the Online Auction System. This module allows users to view auction items, place bids, and track their activities easily.

Features

- View available auction items with details.
- Search items based on name or category.
- Register and login to the system.
- Place bids on auction items.
- View bidding history.
- Check auction results (win/loss status).
- Receive notifications about bids and results.

6.2 SELLER MODULE :

The Seller Module provides an interface for users who want to sell their products through the Online Auction System. This module allows sellers to add items, manage their listings, and monitor the bidding process easily.

Features

- Add new auction items with details (title, price, image, description).
- Update or delete listed items.
- Set auction start and end time.

- View bids placed on their items.
- Monitor highest bid in real-time.
- Manage item listings.
- View auction results and winning bidder details.

6.3 BUYER MODULE :

The Buyer Module provides an interface for users who want to purchase items through the Online Auction System. This module allows buyers to view items, place bids, and track their auction activities easily.

Features

- View available auction items with details.
- Search items based on name or category.
- Register and login to the system.
- Place bids on auction items.
- View current highest bid.
- Track bidding history.
- Check auction results (win/loss status).
- Receive notifications about bid updates and results.

6.4 AUCTION MODULE:

The Auction Module manages the overall auction process in the Online Auction System. It controls the creation, timing, and execution of auctions, ensuring a fair and smooth bidding process.

Features

- Create and manage auction events.
- Set auction start and end time.
- Display items available for auction.
- Allow users to place bids during the auction.
- Update highest bid in real-time.
- Automatically close auction after end time.
- Identify and declare the highest bidder as winner.
- Store auction results for future reference.

6.5 BIDDING MODULE :

The Bidding Module handles the process of placing and managing bids in the Online Auction System. It ensures that bids are recorded correctly and that the highest bid is updated in real-time.

Features

- Allow users to place bids on items.
- Validate bid amount (must be higher than current bid).
- Update highest bid automatically.
- Display current highest bid to users.
- Maintain bidding history for each item.
- Prevent invalid or duplicate bids.
- Notify users about bid status (outbid/winning).
- Ensure fair and secure bidding process

6.6 ADMIN MODULE :

The Admin Module provides control and management functions for the Online Auction System. It allows the administrator to monitor system activities, manage users, and ensure smooth operation of auctions.

Features

- Manage user accounts (add, update, delete users).
- Approve or remove seller listings.
- Monitor all auction activities.
- Manage auction items and categories.
- View and manage bids.
- Handle user complaints and issues

SAMPLE SQL QUERIES

1.INSERT NEW USER

```
INSERT INTO users (email, password)
VALUES ('user@gmail.com', '123456');
```

2. Insert New Item

```
INSERT INTO items (title, starting_price, image)
VALUES ('Laptop', 25000, 'laptop.jpg');
```

3. Place a Bid

```
INSERT INTO bids (item_id, email, bid_amount)
VALUES (1, 'user@gmail.com', 27000);
```

4. View All Items

```
SELECT * FROM items;
```

5. View Bidding History

```
SELECT * FROM bids  
WHERE item_id = 1;
```

6. Get Winner of an Auction

```
SELECT email, bid_amount  
FROM bids  
WHERE item_id = 1  
ORDER BY bid_amount DESC  
LIMIT 1;
```

8. User Login Check

```
SELECT * FROM users  
WHERE email = 'user@gmail.com' AND password = '123456';
```

7. Delete an Item

```
DELETE FROM items  
WHERE item_id = 1;
```

8. Update Item Details

```
UPDATE items  
SET starting_price = 30000  
WHERE item_id = 1;
```

CHAPTER 7

IMPLEMENTATION

7.1 DATABASE CREATION

The database used in the Online Auction System is **auction_db**. It is created using MySQL to store all the data related to users, items, and bids.

SQL

```
CREATE DATABASE auction_db;  
USE auction_db;
```

The database contains multiple tables with proper relationships using primary keys and foreign keys to ensure data integrity and efficient data management.

7.2 SQL OPERATIONS

7.2.1 Insert Operation

Used to add new records into the database.

SQL

```
INSERT INTO users (email, password)  
VALUES ('user@gmail.com', '123456');
```

```
INSERT INTO items (title, starting_price, image)  
VALUES ('Mobile Phone', 15000, 'mobile.jpg');
```

```
INSERT INTO bids (item_id, email, bid_amount)  
VALUES (1, 'user@gmail.com', 16000);
```

7.2.2 Select Operation

Used to retrieve data from the database.

SQL

```
SELECT * FROM users;
```

```
SELECT * FROM items WHERE starting_price > 10000;
```

```
SELECT * FROM bids WHERE item_id = 1;
```

7.2.3 Update Operation

Used to modify existing records.

SQL

```
UPDATE items  
SET starting_price = 18000  
WHERE item_id = 1;
```

```
UPDATE users  
SET password = 'newpassword'  
WHERE id = 1;
```

7.2.4 Delete Operation

Used to remove records from the database.

SQL

```
DELETE FROM bids WHERE bid_id = 2;
```

```
DELETE FROM items WHERE item_id = 1;
```

7.3 CONSTRAINTS

In a database, constraints are rules applied to columns or tables to ensure the accuracy, integrity, and reliability of stored data. Constraints enforce business rules at the database level and help prevent invalid or inconsistent data entry.

In the context of an Online Auction System, constraints are important because the system manages critical data such as user details, auction items, bids, and transactions. Without constraints, there could be duplicate users, invalid bids, incorrect pricing, or unauthorized actions, leading to system errors and unfair auctions.

7.3.1 Primary Key

A Primary Key is used to uniquely identify each record in a table. It cannot contain NULL values and must be unique.

Example:**SQL**

```
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    email VARCHAR(255) NOT NULL,
    password VARCHAR(255) NOT NULL
);
```

Explanation:

- id is the primary key
- Each user will have a unique ID
- No duplicate or NULL values are allowed

7.3.2 Foreign Key

A Foreign Key is used to create a relationship between two tables. It ensures that the value in one table matches the primary key in another table.

Example:**SQL**

```
CREATE TABLE bids (  
    bid_id INT AUTO_INCREMENT PRIMARY KEY,  
    item_id INT,  
    email VARCHAR(255),  
    bid_amount DECIMAL(10,2),  
    FOREIGN KEY (item_id) REFERENCES items(item_id)  
);
```

Explanation:

- item_id in bids table refers to item_id in items table
- Ensures only valid item IDs are used
- Maintains referential integrity

7.3.3 Not Null

The NOT NULL constraint ensures that a column cannot have empty (NULL) values.

Example:

SQL

```
CREATE TABLE items (  
    item_id INT PRIMARY KEY,  
    title VARCHAR(255) NOT NULL,  
    starting_price DECIMAL(10,2) NOT NULL  
);
```

7.3.4 Unique

The UNIQUE constraint ensures that all values in a column are different.

Example:

SQL

```
CREATE TABLE users (  
    id INT PRIMARY KEY,  
    email 78VARCHAR(255) UNIQUE  
);
```


CHAPTER 8

CONCLUSION

The Online Auction System developed in this project successfully demonstrates the effective use of database management concepts in handling real-world e-commerce activities. The system provides an organized and efficient way to manage users, auction items, and bidding processes.

The system improves interaction between buyers and sellers, making the auction process faster, transparent, and more reliable. Overall, the project fulfills its objectives and provides a scalable and user-friendly solution for online auctions.

KEY ACHIEVEMENTS

The following achievements were accomplished during the development of the system:

Structured Database Design – A well-organized relational database with proper tables and relationships

Efficient Data Management – Easy storage, retrieval, and updating of user, item, and bid records

Real-time Bidding System – Accurate tracking of bids and highest bidder

Reduction of Manual Errors – Automated processes minimize human mistakes

Implementation of Constraints – Ensures data integrity using primary key, foreign key, and other constraints

- **User-Friendly Interface** – Simple and easy-to-use system design

FUTURE ENHANCEMENTS

The system can be further improved by adding the following features:

User Authentication System – Role-based login for admin, buyer, and seller

Online Payment Integration – Secure payment methods for completed auctions

Email/SMS Notifications – Alerts for bid updates and auction results

Mobile Application – Develop Android/iOS app for easy access

Advanced Search & Filters – Improve item discovery experience

Security Enhancements – Stronger encryption and fraud detection

CONCLUSION

The Online Auction System developed in this project provides an efficient and reliable solution for managing online auction activities. Overall reduces human errors, and improves user experience. This project demonstrates the practical application of database management concepts and serves as a strong foundation for future improvements in e-commerce systems.

APPENDIX A: SOURCE CODE STRUCTURE

```
. * {  
  
    margin: 0;  
  
    padding: 0;  
  
    box-sizing: border-box;  
  
    font-family: 'Poppins', sans-serif;  
  
}  
  
body {  
  
    min-height: 100vh;  
  
    background: linear-gradient(135deg, #667eea, #764ba2);  
  
    display: flex;  
  
    justify-content: center;  
  
    align-items: center;  
  
}  
  
/* FORM CONTAINER */  
  
.form-container {
```

```
width: 350px;

padding: 30px;

text-align: center;

background: rgba(255, 255, 255, 0.15);

border-radius: 15px;

backdrop-filter: blur(12px);

box-shadow: 0 8px 32px rgba(0,0,0,0.3);

}

/* ITEMS GRID */

.items-container {

    display: flex;

    flex-wrap: wrap;

    justify-content: center;

    gap: 20px;

    padding: 20px;

}

/* ITEM BOX */

.item-box {
```

```
width: 250px;

background: white;

color: black;

padding: 15px;

border-radius: 12px;

box-shadow: 0px 4px 10px rgba(0,0,0,0.2);

text-align: center;

transition: 0.3s;

}

.item-box:hover {

    transform: scale(1.05);

}

/* IMAGE */

.item-box img {

    width: 100%;

    height: 150px;

    object-fit: cover;

    border-radius: 10px;
```

```
}
```

```
/* HEADINGS */
```

```
h1 {
```

```
    color: white;
```

```
    margin-bottom: 20px;
```

```
}
```

```
/* BUTTON */
```

```
button {
```

```
    width: 100%;
```

```
    padding: 10px;
```

```
    margin-top: 10px;
```

```
    border: none;
```

```
    border-radius: 8px;
```

```
    background: #333;
```

```
    color: white;
```

```
    font-weight: bold;
```

```
    cursor: pointer;
```

```
}
```

```
button:hover {
```

```
    background: #555;
```

```
}
```

```
/* INPUT */
```

```
input, select {
```

```
    width: 100%;
```

```
    padding: 10px;
```

```
    margin: 10px 0;
```

```
    border-radius: 8px;
```

```
    border: none;
```

```
<?php
```

```
$conn = mysqli_connect("localhost", "root", "", "auction_db");
```

```
if(!$conn){
```

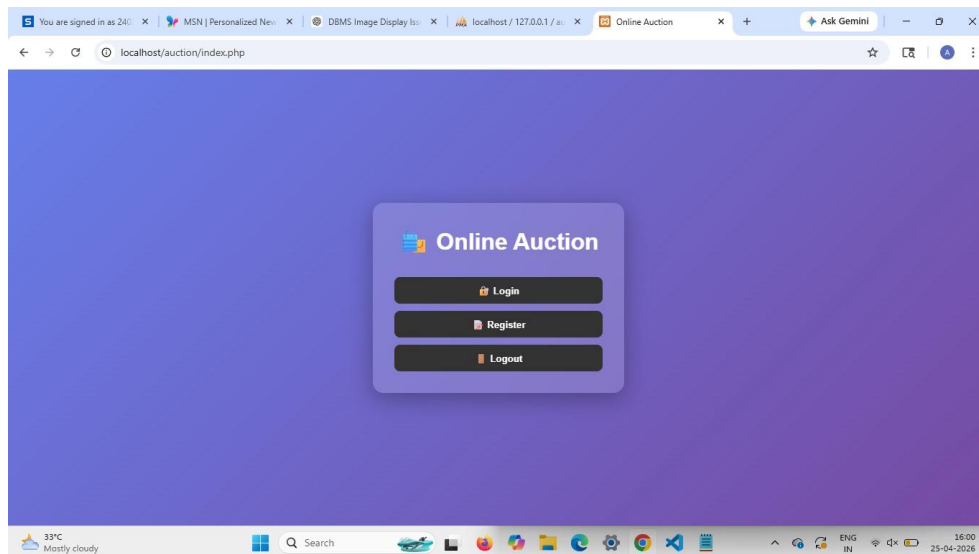
```
    die("Connection Failed");
```

```
}
```

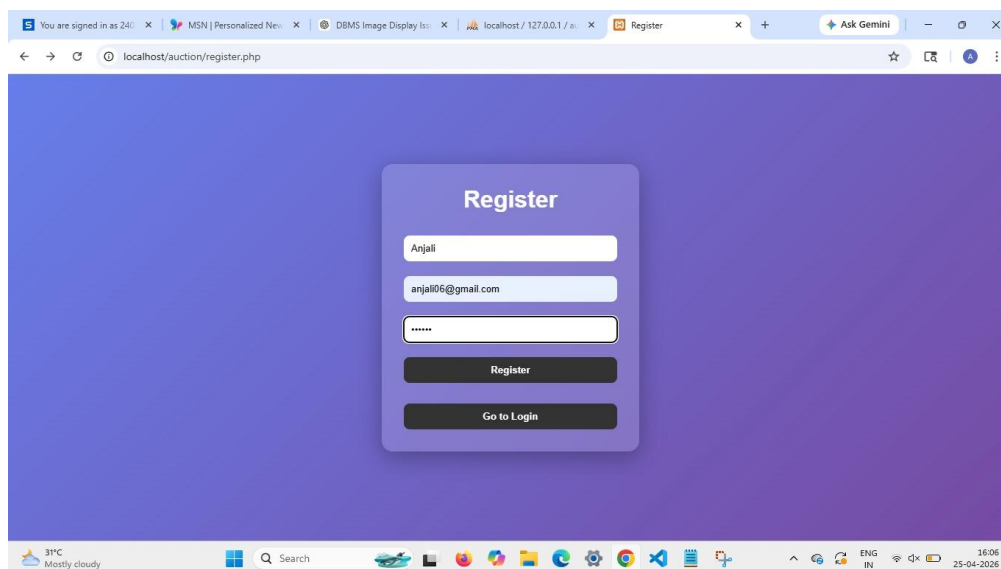
```
?>}
```

APPENDIX A SOURCE CODE

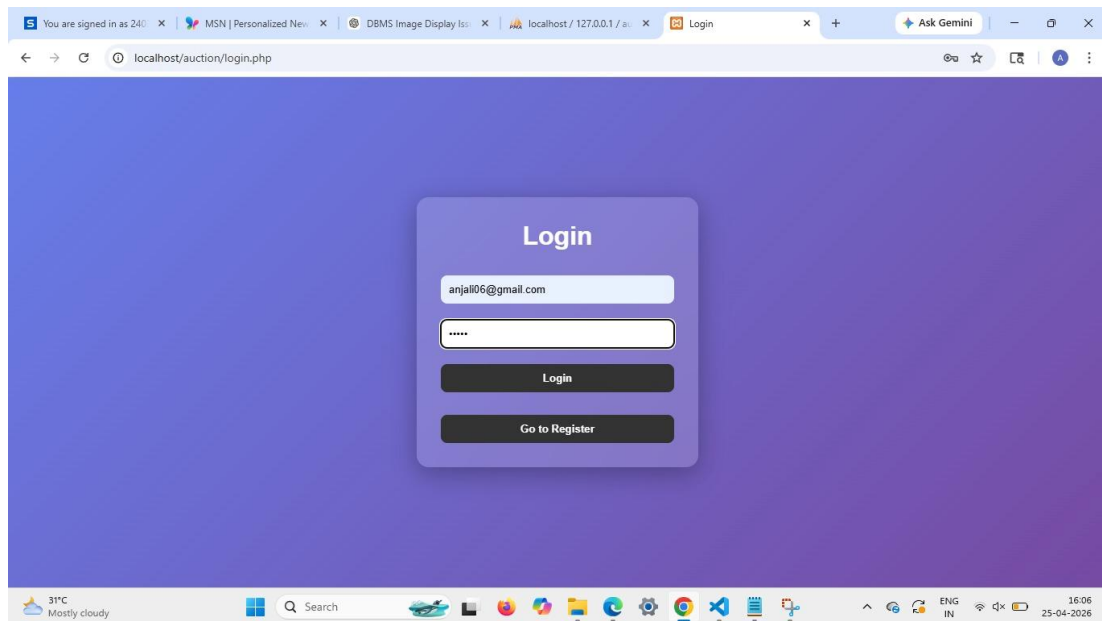
DASHBOARD:



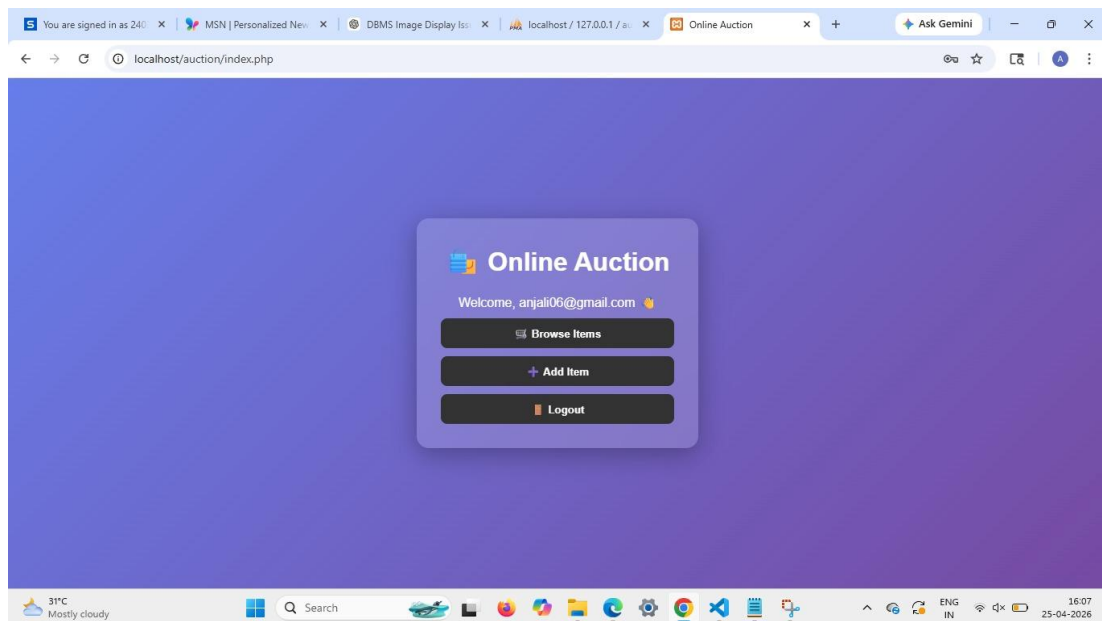
REGISTER:



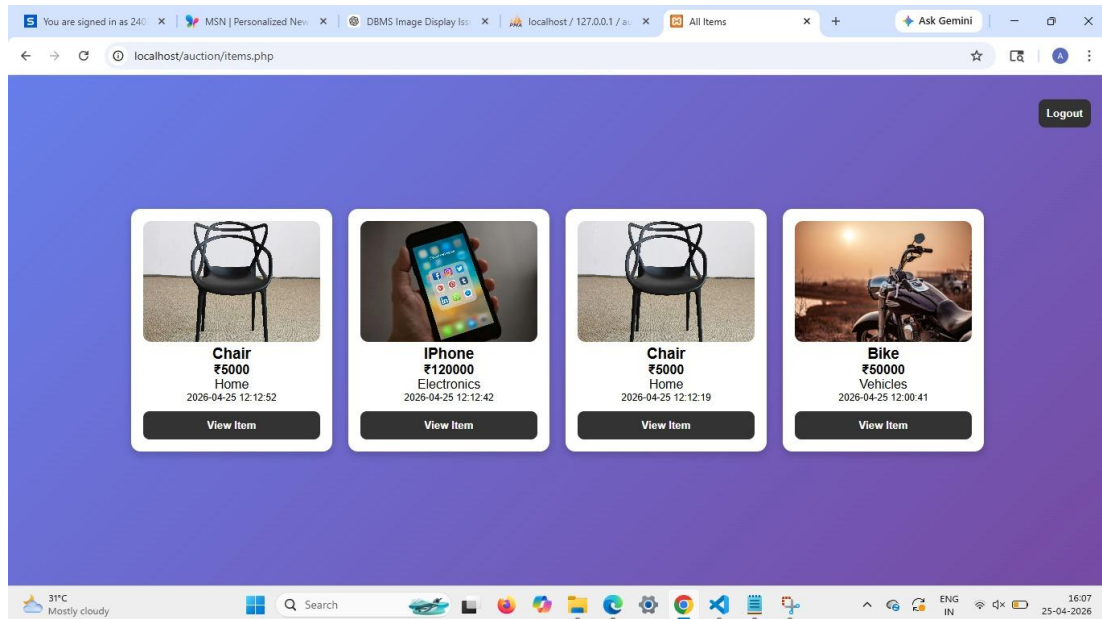
LOGIN:



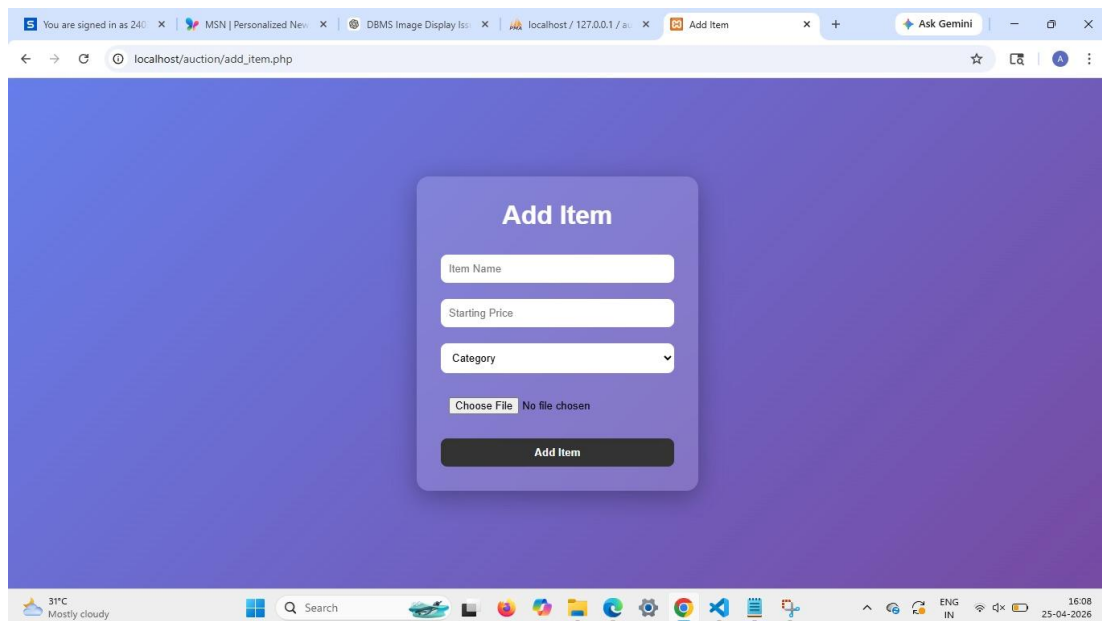
AUCTION SYSTEM:



AVAILABLE ITEMS:



ADD ITEM:



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9. Wang et al. (2002), Auction Design and Online Market Efficiency, International Journal of Electronic Commerce.
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